

Food Delivery Time Prediction

1. Project Overview

This project analyzes food delivery transactional data to understand delivery performance across different restaurants, locations, vehicle types, and time periods. The goal is to identify factors affecting delivery time, uncover delay patterns, and predict estimated delivery time to improve operational efficiency and customer satisfaction.

2. Dataset Summary

- **Rows:** [50000]
- **Columns:** [24]
- **Key Features:**
 - **Order details:** Order ID, restaurant, order date/time, food category
 - **Delivery details:** Delivery distance, pickup time, delivery time, vehicle type
 - **External factors:** Traffic conditions, weather, location
 - **Customer details:** Customer location, order frequency, ratings
 - **Performance metrics:** Preparation time, delivery time, delay status
- **Target Variable:** Time to Deliver
 - **Missing Data:** Checked and handled missing values during data preprocessing.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.

[2]:

	Order_ID	Order_Date	Order_Hour	Day_of_Week	Is_Weekend	Is_Festival	Weather	Pickup_Zone	Dropoff_Zone	Vehicle_Type	...	Order_Items	Restaurant_Load
0	ORD000001	2025-10-28	8	Tuesday	0	0	Clear	CBD	Commercial	Scooter	...	1	Medium
1	ORD000002	2025-09-26	12	Friday	0	0	Rain	CBD	CBD	Scooter	...	3	Medium
2	ORD000003	2025-01-26	18	Sunday	1	0	Clear	Residential	Residential	Scooter	...	1	Low
3	ORD000004	2025-11-19	12	Wednesday	0	0	Cloudy	Residential	Industrial	Bike	...	3	Medium
4	ORD000005	2025-08-04	18	Monday	0	0	Rain	Industrial	Residential	Bike	...	1	Medium

5 rows × 24 columns

- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```

<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype  
---  -
0   Order_ID                             50000 non-null  str    
1   Order_Date                           50000 non-null  str    
2   Order_Hour                           50000 non-null  int64  
3   Day_of_Week                           50000 non-null  str    
4   Is_Weekend                           50000 non-null  int64  
5   Is_Festival                           50000 non-null  int64  
6   Weather                               50000 non-null  str    
7   Pickup_Zone                           50000 non-null  str    
8   Dropoff_Zone                          50000 non-null  str    
9   Vehicle_Type                          50000 non-null  str    
10  Rider_Experience_Years                 50000 non-null  float64
11  Rider_Rating                           50000 non-null  float64
12  Restaurant_Rating                      50000 non-null  float64
13  Cuisine_Type                           50000 non-null  str    
14  Order_Items                           50000 non-null  int64  
15  Restaurant_Load                         50000 non-null  str    
16  Preparation_Time_Min                   50000 non-null  int64  
17  Road_Distance_km                       50000 non-null  float64
18  Delivery_Distance_Category             50000 non-null  str    
19  Traffic_Level                          50000 non-null  str    
20  Number_of_Signals                      50000 non-null  int64  
21  Average_Speed_kmph                     50000 non-null  float64
22  Delivery_Priority                      50000 non-null  str    
23  Time_taken_min                         50000 non-null  int64  
dtypes: float64(5), int64(7), str(12)
memory usage: 13.1 MB

```

[5]:

	Order_ID	Order_Date	Order_Hour	Day_of_Week	Is_Weekend	Is_Festival	Weather	Pickup_Zone	Dropoff_Zone	Vehicle_Type	...	Order_Items	Resta
count	50000	50000	50000.000000	50000	50000.000000	50000.000000	50000	50000	50000	50000	...	50000.000000	
unique	50000	365	NaN	7	NaN	NaN	5	5	5	4	...	NaN	
top	ORD000001	2025-02-12	NaN	Tuesday	NaN	NaN	Clear	Residential	Residential	Bike	...	NaN	
freq	1	170	NaN	7264	NaN	NaN	23992	15786	16106	22943	...	NaN	
mean	NaN	NaN	13.847700	NaN	0.283600	0.049100	NaN	NaN	NaN	NaN	...	2.652740	
std	NaN	NaN	5.525291	NaN	0.450749	0.216079	NaN	NaN	NaN	NaN	...	1.590359	
min	NaN	NaN	0.000000	NaN	0.000000	0.000000	NaN	NaN	NaN	NaN	...	1.000000	
25%	NaN	NaN	10.000000	NaN	0.000000	0.000000	NaN	NaN	NaN	NaN	...	1.000000	
50%	NaN	NaN	14.000000	NaN	0.000000	0.000000	NaN	NaN	NaN	NaN	...	2.000000	
75%	NaN	NaN	19.000000	NaN	1.000000	0.000000	NaN	NaN	NaN	NaN	...	4.000000	
max	NaN	NaN	23.000000	NaN	1.000000	1.000000	NaN	NaN	NaN	NaN	...	7.000000	

11 rows × 24 columns

For your **Food Delivery Time Prediction** project, you can write the data preprocessing section in the same format:

- **Missing Data Handling:** Checked for null values and handled missing values using appropriate imputation techniques.
- **Column Standardization:** Renamed columns to **snake_case** for better readability and consistency.
- **Feature Engineering:**
 - Created vehicle_type to categorize delivery vehicles.
 - Created order items and delivery_time in different traffic levels.
 - Created cuisine_type and food preparation_time.
 - Created vehicle_type and ride_ratings.
 - Created restaurant and dropoff locations.
- **Data Consistency Check:** Verified duplicate records, invalid delivery times, and inconsistent values across restaurant, pickup, and delivery data.
- **Database Integration:** Connected Python to **PostgreSQL** and loaded the cleaned DataFrame into the database for SQL analysis and Power BI reporting.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Number of orders for each restaurant** – Displays number of orders for each restaurant.

	pickup_zone text	sum numeric
1	Suburban	17386
2	Residential	41571
3	CBD	29085
4	Industrial	19501
5	Commercial	25094

2. **Average delivery time for each vehicle type** – Calculates average delivery time for each vehicle type.

	vehicle_type text	round numeric
1	Bike	71.95
2	Scooter	82.36
3	Bicycle	130.40
4	Electric Scoot...	89.05

3. **Average delivery time for each weather condition** – Calculates average delivery time for each weather condition.

	weather text	round numeric
1	Storm	114.45
2	Rain	98.86
3	Clear	77.30
4	Cloudy	79.02
5	Fog	84.15

4. **Percentage of orders delivered in less than 30 minutes** – Calculates percentage of orders delivered within 30 minutes to customers.

	delivered_in_30_min numeric
1	1.61

5. **Top 5 restaurants with highest average delivery time** – Displays top 5 restaurants with highest average delivery time.

	pickup_zone text	round numeric
1	CBD	93.84
2	Suburban	83.27
3	Commercial	83.26
4	Residential	80.27
5	Industrial	77.91

6. **Top 5 Pickup Zones with the most orders in rain** – Displays most orders delivered from pickup zones in rain.

	pickup_zone text	sum numeric
1	Residential	7400
2	CBD	5567
3	Commercial	4447
4	Industrial	3379
5	Suburban	3110

7. **Top 5 Fastest and slowest deliveries from each pickup zone** – Displays Top 5 Fastest and slowest deliveries from each pickup zones.

	pickup_zone text	fast_delivery bigint	slow_delivery bigint
1	Suburban	12	180
2	Residential	10	180
3	CBD	18	180
4	Industrial	14	180
5	Commercial	14	180

8. **Peak ordering hour with number of orders** – Calculates the number of orders in peak ordering hour.

	order_hour bigint	count bigint
1	19	4773

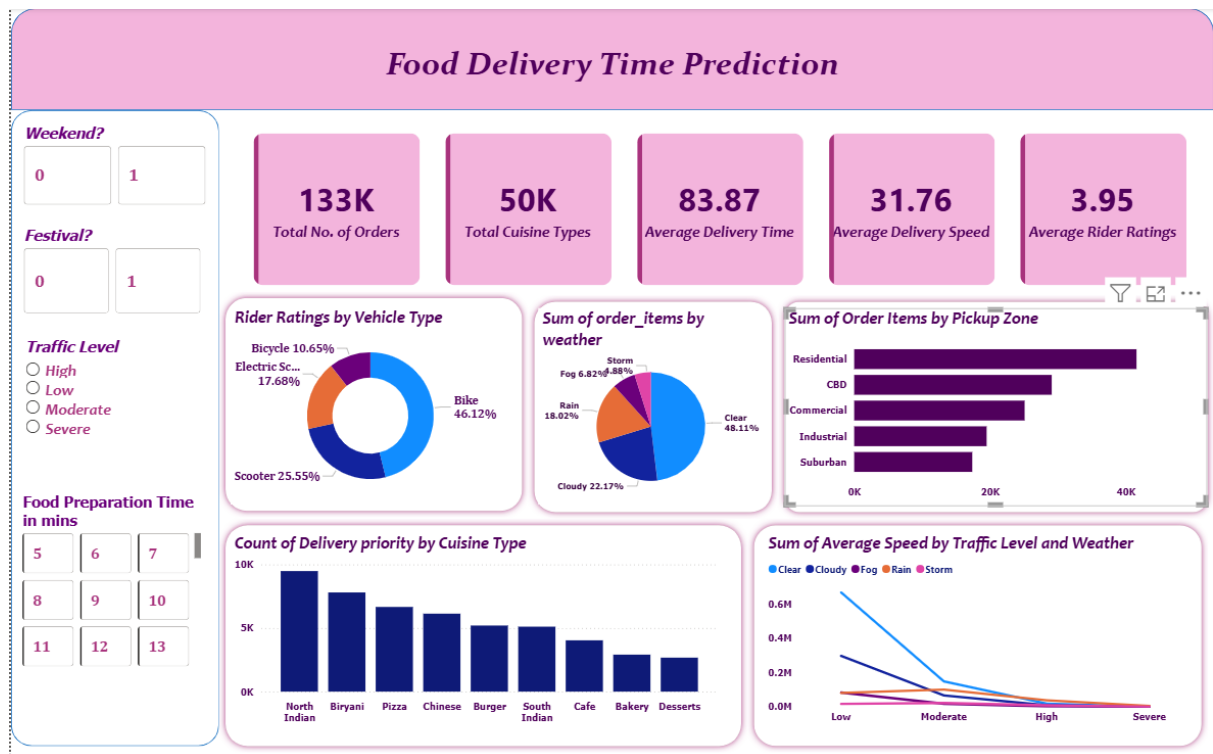
9. **Number of delayed orders (>45 minutes)** – Calculates number of orders delayed i.e, delivered in more than 45 minutes.

	pickup_zone text	delayed_orders bigint
1	Suburban	5741
2	Residential	13800
3	CBD	10408
4	Industrial	6105
5	Commercial	8469

10. **Number of orders delivered based on traffic level** – Calculates the count of orders delivered based on different traffic levels.

	traffic_level text	count bigint
1	Low	32567
2	Moderate	13165
3	High	3744
4	Severe	524

5. Dashboard in Power BI



6. Business Recommendations

- **Optimize Delivery Operations** – Allocate delivery partners based on order volume, location, and peak hours.
- **Reduce Delivery Delays** – Identify major delay factors such as traffic, weather, distance, and restaurant preparation time.
- **Improve Restaurant Performance** – Monitor restaurant preparation times and work with high-delay restaurants to improve efficiency.
- **Vehicle Optimization** – Assign suitable vehicle types based on delivery distance and traffic conditions.
- **Peak-Hour Planning** – Increase delivery partner availability during high-demand periods to reduce delivery time.
- **Customer Experience** – Provide accurate estimated delivery times and use delivery performance insights to improve customer satisfaction.